BARSHAN GHOSH

Computer Science Graduate | Software & IT Systems | Applied AI | Technology Education

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Portfolio | All GitHub Repositories | GitHub Profile | LinkedIn

PROFESSIONAL SUMMARY

Computer Science and Engineering graduate (CGPA 3.83/4.00) with experience across IT operations and consulting, web/software projects, applied AI/ML learning, digital marketing, and technology education. Builds practical projects using modern web technologies and database-backed systems, with a particular focus on data integrity, workflow automation, and usable tools for real-world problems. Also experienced in school IT coordination, computer instruction, and private tutoring.

TECHNICAL SKILLS

- **Web & Software:** Next.js, React, Astro, TypeScript, JavaScript, HTML/CSS, Tailwind CSS, Node.js, REST APIs
- **Databases & Platforms:** PostgreSQL, Supabase, MySQL, Firebase, SQL Server, WordPress, Shopify
- **AI, Data & Testing:** Python, Weka, TensorFlow/Keras fundamentals, Power BI, Vitest, Playwright, Kaggle
- **Systems & Tools:** Git/GitHub, Vercel, Linux, Windows, hardware and network troubleshooting, Docker basics
- **Digital & Education Tools:** SEO, GA4, Google Tag Manager, Meta Pixel, Google Search Console, Canva, Google Workspace, MS Office, Google Classroom, Zoom, Microsoft Teams
- **Additional Programming**: C, C++, Java; PHP (basic reading and modification)

SELECTED PROJECTS

- **Money Calculator | Next.js, TypeScript, Supabase, PostgreSQL, Vitest, Playwright**
  In-progress personal-finance application built around ledger-backed records. Includes transaction correction/reversal history, account balances, lending and borrowing lifecycle, tuition payments, asset tracking, budgeting, reporting, and financial forecasts. Repository includes unit and database-integration testing for key financial invariants.
  View source repository
- **ASTA Store | Astro, TypeScript, Supabase**
  E-commerce storefront for handmade leather bags with product browsing, cart management, checkout flow, Supabase authentication, and an administrative product-management area.
  View source repository
- **Mount Royal Academy Website | Astro, TypeScript, Supabase**
  School website project with admissions/application flow, academic calendar, notices, gallery management, and authenticated content administration.
  View source repository
- **999 National Emergency Service | C#, Windows Forms, SQL Server**
  Academic desktop application concept for citizen registration, emergency-service request entry with time/location details, and administrative request monitoring. Not a live emergency dispatch service.
  View source repository
- **Blockchain Internet of Things (bIoT) Thesis | Research**
  Undergraduate literature review and proposed architecture exploring blockchain-based security for healthcare IoT, including authentication, data integrity, remote patient monitoring, and medical-data privacy.
  View source repository
- **Student Performance Dashboard | Analytics & Data Visualization**
  Dashboard concept for presenting student results, attendance, and classroom-performance indicators using illustrative sample data; not connected to a live school database.
  View source repository
- **Digital Coaching Spreadsheet Automation | Google Apps Script**
  Small workflow automation that adds a missing current-date column to an Attendance sheet and a missing current-month column to a Payment sheet, avoiding duplicate headers.
  View source repository

PROFESSIONAL EXPERIENCE

- **IT Coordinator | Computer Teacher | O-Level Physics Guest Lecturer** | *Jun 2025 – Present*
  Mount Royal Academy — Narayanganj, Bangladesh
  • Support day-to-day school IT infrastructure and technology operations.
  • Teach computer/ICT concepts to students in Grades 3–7.
  • Deliver guest instruction in O-Level Physics.
- **Proxy ICT Teacher** | *Apr 2025 – May 2025*
  Heritage School — Narayanganj, Bangladesh
  • Covered ICT classes during regular teachers' absences and maintained continuity with planned lessons.
  • Led classroom technology activities and supported student learning.

- **Digital Marketer (Remote)** | *Oct 2024 – Jan 2025*
  Edupy Academy — Dhaka, Bangladesh
  • Supported digital visibility, social-media outreach, and promotional content across online channels.
- **IT Associate (Hybrid)** | *Jan 2023 – Sep 2024*
  CNBDEXIM — Dhaka, Bangladesh
  • Maintained day-to-day IT systems and assisted with hardware support and network troubleshooting.
  • Provided on-site and remote technical assistance for business operations.
- **IT Consultant (Remote)** | *Aug 2022 – Oct 2024*
  Digital Fire Safety Solution Ltd — Dhaka, Bangladesh
  • Provided remote technical consulting and troubleshooting for operational IT issues.
  • Reviewed IT needs and recommended practical improvements to system reliability and efficiency.
- **AI Engineer Intern (Remote)** | *Aug 2022 – Jan 2023*
  Teachnook — India
  • Participated in AI/ML learning and project activities involving model-building, training, and testing workflows.
  • Completed a cancer-prediction learning exercise using a Kaggle dataset and tools including Weka, Excel, and Google Colab.

EDUCATION

- **B.Sc. in Computer Science and Engineering** | 2019 – 2023
  American International University-Bangladesh (AIUB) — CGPA: 3.83/4.00; completed in 3 years
- **Higher Secondary Certificate (Science)** | 2017 – 2019
  Government Tolaram College — GPA: 3.92/5.00
- **Secondary School Certificate (Science)** | 2014 – 2017
  Narayanganj High School & College — GPA: 5.00/5.00

PRIVATE TUTORING EXPERIENCE

- O-Level / Edexcel (Jun 2026) — Physics, ICT, Computer Science, Mathematics B, Further Pure Mathematics
- HSC Candidate (Science) (Oct 2024) — ICT, English, Bengali
- SSC Candidate (Arts) (Jan 2023) — ICT, English, General Mathematics, General Science
- SSC Candidate (Science) (Jan 2023) — Physics, Chemistry, Higher Mathematics, Biology

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

• Operating System — Coursera | Verify credential
• AI for Everyone — Coursera | Verify credential
• Python for Everybody — Coursera Specialization | Verify credential
• IT Fundamentals for Cybersecurity — Coursera Specialization | Verify credential
• MS Excel: Beginner to Advanced | Verify credential
• MS PowerPoint Advanced Course | Verify credential
• Professional Digital Marketing (A+) — Creative IT | Verify credential
• Digital Marketing Business Workshop | Verify credential

LANGUAGES

Bengali (native) | English | Hindi